Massey University
College of Sciences

161.304: Computer Laboratory 9

Exercise 1: Professor Kolmogorov Plays Cricket
Professor Kolmogorov is a keen cricketer. He has noted that his batting performances tend to go through periods of good and bad form, and so decides to model them using a Markov chain. Specifically, he categorizes each of his innings as follows:

Category	Definition
1	Disaster: made 5 runs or less.
2	Poor: made between 6 and 15 runs.
3	Adequate: made between 16 and 29 runs.
4	Good: made between 30 and 49 runs.
5	Excellent: made 50 or more runs.

The transition matrix for this chain is

$$P = \begin{bmatrix} 0.4 & 0.2 & 0.2 & 0.1 & 0.1 \\ 0.3 & 0.3 & 0.2 & 0.1 & 0.1 \\ 0.2 & 0.2 & 0.3 & 0.2 & 0.1 \\ 0.1 & 0.2 & 0.3 & 0.2 & 0.2 \\ 0.1 & 0.1 & 0.2 & 0.4 & 0.2 \end{bmatrix}$$

1. Create a matrix `P` in R for the transition matrix.
2. Remember that the transition matrix is stochastic: i.e. the row sums are all one. Check that this is the case with the command

   ```
   rowSums(P)
   ```

   If any of the row sums are not equal to one, then you have made an error when transcribing P. Find and correct it.
3. Professor Kolmogorov starts the season in disastrous form; i.e. $Y_0 = 1$. Find the probability that he is in excellent form at innings $t = 5$ (i.e. find $\mathsf{P}(Y_5 = 5|Y_0 = 1)$).
4. In innings $t = 10$, Professor Kolmogorov scores 20 runs. Find the probability that he will score 30 or more runs in innings $t = 13$.

Exercise 2: A Loopy Exercise
Write loops in R to create vectors containing the first 20 elements of the following sequences. To maintain the connection with Markov chains, we use the notation y_t for the tth element when describing the sequences below.

1. The sequence $\{1, 2, 5, 14, \ldots\}$ defined by $y_1 = 1$ (the starting value) and then $y_{t+1} = 3y_t - 1$.
2. The Fibonacci numbers, $\{0, 1, 1, 2, 3, 5, 8, 13 \ldots\}$, where $y_{t+1} = y_t + y_{t-1}$. The first two entries, $y_1 = 0$ and $y_2 = 1$, are considered to be fixed starting values.

Exercise 3: Simulating Professor Kolmogorov's Cricket Season
Assuming that he starts in disastrous form, so $Y_0 = 1$, write R code to simulate Professor Kolmogorov's batting performance over the next $n = 20$ innings using the Markov chain model from Exercise 1. Produce a plot of y_t against t to display the results.

For how innings many of these 20 innings can Professor Kolmogorov expect to be excellent? To answer this question you will need to produce a large number of different simulations of the season, and then look at the average results. Specifically, to create 1000 separate simulations, you will need to write a *double loop*, of the form

```
for ( i in 1:1000 ){
.....
      for (tt in 1:20){
      .....
      }
results[i] <- .....
}
```

where `results` is a vector of length 1000 for which the ith element will store the number of excellent innings in the ith simulation.

The first row of dots will need to be replaced by any necessary set up for each inner loop. For the final set of dots (assigning a value of `results[i]`), you will need to extract from a vector of simulated states the number which are equal to 5. Suppose that the simulations are held in a vector `y`. Then `sum(y==5)` returns the number of elements of `y` that are equal to 5. To understand why, recall that `y==5` is a logical comparison, and so will return a sequence of `T` (for each element of `y` that equals 5) and `F` (for each element that does not). When applying a numerical function like `sum`, R coerces the logical vector into a binary vector (1 replaces `T` and 0 replaces `F`) and so the sum of these is the number of elements of `y` that equal 5.

Consult the Comments and Solutions sheet if you get completely stuck, but make sure that you have a good try at figuring out your own solution before doing so.

Finally, note that there was no compulsion for me to indent the code for the inner loop in the R syntax given above. Nonetheless, many programmers find it helpful to do so in order to keep track of the structure of their code.

Exercise 4: Calculating Stationary Distributions
Compute the stationary distributions for Markov chains with state space $S = \{1, 2, 3\}$ the following transition matrices:

$$1.\quad P = \begin{bmatrix} 0.3 & 0.6 & 0.1 \\ 0.5 & 0.2 & 0.3 \\ 0.3 & 0.3 & 0.4 \end{bmatrix} \qquad \text{and} \qquad 2.\quad P = \begin{bmatrix} 0.9998 & 0.0001 & 0.0001 \\ 0.0005 & 0.4995 & 0.5000 \\ 0.0001 & 0.2999 & 0.7000 \end{bmatrix}.$$

Use both methods described in Example 43 of lectures or Example 4.16 of notes, and compare the results. For the 'raise P to a high power' method, try a power of 100 in the first instance.

Exercise 5: Markov Chain Convergence
Assuming that the chains start in state 1, (so $\boldsymbol{\pi}^{(0)} = (1, 0, 0)$), simulate both Markov chains from the previous Exercise for $n = 100,000$ iterations (after the initial state). Store the results as vectors `y1` and `y2` respectively. Produce lines plots of the simulated Markov chains for the time range $t = 0$ to 100 (i.e. plot y_t against t in each case, with values connected by lines rather than plotted as individual points.)

Both the Markov chains are regular, so we know that in each case the distribution $\boldsymbol{\pi}^{(t)}$ will eventually settle down (as t becomes large) to the stationary distribution of the chain in question. This should be reflected in your simulations, in that the proportions of time spent in each state should approximately match the stationary distribution, once the Markov chain has converged.

You can compute the proportion of times sent in each state for a simulated chain `y`

```
mc.props <- tabulate(y,nbins=3)/length(y)
```

where `tabulate(y,nbins=m)` provides the frequencies with which elements of the argument `y` occur, for which it is assumed that `y` takes values in the set $\{1, 2, \ldots, m\}$. Apply this methodology to both chains, and then compare the proportions with the stationary distributions computed in Exercise 1.

You should find that the proportions for the first chain match the stationary distribution probabilities quite closely. For the second chain it is likely that the proportions will be far less close to the stationary probabilities, although your results will of course depend on the setting of the random number seed. Try repeating the process for the second chain using $n = 1,000,000$. Does this help?